Mohammad VazirPanah

Embedded & Real-Time Systems Engineer

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LinkedIn · Website · GitHub

Summary

Embedded and real-time systems engineer with a research background in heterogeneous and RISC-V architectures. Hands-on with C/C++ across embedded CPUs, GPUs, and FPGAs; real-time software (FreeRTOS); hardware/software co-design; and compiler and performance evaluation, supported by reproducible Linux toolchains (Git, Docker, CI, CMake) and IEEE/ACM publications.

Technical Skills

Programming	C, C++, Python, Assembly
Embedded & Real-Time	FreeRTOS, RISC-V, FPGA, VHDL, Verilog
Parallel & Heterogeneous	SYCL, OpenMP, OpenCL, CUDA
Modeling & Verification	SystemC, NuSMV, UPPAAL
Simulation & EDA	Gem5, QEMU, OVPsim, Quartus, ModelSim, MATLAB-Simulink
Tools & DevOps	Git, Docker, Linux, CMake, CI
Robotics	ROS, Gazebo, sensor integration
Certifications	TinyML — Harvard University, edX

Professional Experience

Research Fellow *2023 – 2026*
University of Salerno — ISISLab, Salerno, Italy

- Developed and deployed portable real-time applications across heterogeneous embedded CPU/GPU/RISC-V platforms (C/C++, SYCL, OpenMP, OpenCL) in the EU-funded LibreRT project
- Benchmarked GCC, LLVM, and LLVM-EPI auto-vectorization on RISC-V vector hardware across 151 TSVC loops and seven real-world applications; characterized per-pattern compiler limitations and showed LMUL tuning yields up to 3× speedup (IEEE PDP 2025, 2nd author)
- Performed bring-up and evaluation of RISC-V/SoC boards; built reproducible build and test pipelines with Git, Docker, CI, and CMake on Linux
- Carried out hardware/software co-design and performance analysis across embedded and HPC targets

Selected Projects

FIR Filter & Tiny YOLO v1 on FPGA *2019*
M.Sc. System-on-Chip — DE0-Nano / DE10 Intel FPGA

- Implemented a FIR filter IP core in Quartus with an Avalon-MM interface
- Deployed a real-time Tiny YOLO v1 (CNN) object-detection inference pipeline on FPGA under tight resource constraints

Self-Aware CPS Modeling Framework *2022*
M.Sc. Thesis — ForSyDe-SystemC, ROS/Gazebo co-simulation

- Built a self-aware cyber-physical system model and contributed extensions to the ForSyDe-SystemC library

Railroad Control System Verification *2020*
M.Sc. Project — NuSMV

- Modeled and formally verified safety and liveness properties of a railroad control system via model checking

RISC-V Benchmark Suite *2025*
Open-source, GitHub

- Implemented an open-source RISC-V/RVV benchmark suite and evaluated GCC/LLVM auto-vectorization across compiler versions

Education

M.Sc. Computer Engineering — Computer Systems Architecture *2019 – 2022*
Shahid Beheshti University, Tehran, Iran

- GPA 3.79 / 4.00 · Thesis: *Modeling Self-Aware Cyber-Physical Systems with SystemC*

B.Sc. Computer Engineering — Hardware Engineering *2012 – 2016*
Birjand University, Birjand, Iran

- GPA 3.40 / 4.00

Selected Publications

- L. Carpentieri, **M. VazirPanah**, B. Cosenza. *A Performance Analysis of Autovectorization on RVV RISC-V Boards.* IEEE PDP 2025.
- B. Cosenza et al. (incl. **M. VazirPanah**). *LibreRT: Portable Heterogeneous Real-Time Programming for the Embedded Computing Continuum.* ACM Computing Frontiers 2026.
- **M. VazirPanah**, S.-H. Attarzadeh-Niaki, A. Salimi-Badr. *ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design.* IEEE CSICC 2022.
- S.-H. Attarzadeh-Niaki, **M. VazirPanah**. *System-Level Modeling of Dynamic Applications with Scenario-Aware Dataflow Graphs.* JICSE 2023.

Honours

- National R&D Research Grant Winner — MCI R&D Center, Iran (2020)
- Top 15% B.Sc. Computer Engineering — Birjand University (2016)

Languages

Persian (Native) · English (Proficient) · Italian (Basic)

References available on request.